



MASTER-KIDS · OLYMPIAD PRACTICE SERIES · CLASS 4

Mathematics Olympiad Worksheet 29

Level 2 — Intermediate · Logical Reasoning II

Class	Class 4 (Primary · Stage 2)	Subject	Mathematics
Level	Level 2 — Intermediate	Topic Focus	Logical Reasoning II — Alphabet, Combinations & Puzzles
Questions	20	Time Allowed	35 minutes
Total Marks	20	Worksheet No.	29 of 100
Curriculum	CBSE · ICSE · State Boards (board-agnostic) · NEP 2020-aligned		

About Master-Kids

Master-Kids is India's cradle-to-career learning platform — one platform that grows with a child from Pre-Nursery all the way to college, turning everyday effort into visible progress for parents and a rewarding game of streaks, XP and badges for children. This Olympiad practice series is part of the Master-Kids Primary School programme (Stage 2, Class 1–5). It is board-agnostic — suited to CBSE, ICSE and State Board children alike — and aligned with the NEP 2020 focus on foundational numeracy, science thinking and language.

About This Worksheet

Worksheet 29 covers the puzzle side of the IMO reasoning section: alphabet position tests, counting possible combinations (outfits, coin flips, seat arrangements), simple family-relation chains, repeating-pattern positions using remainders, and the famous two-mothers-two-daughters riddle that teaches careful reading.

Learning Objectives

- Find midway letters and count letters between positions in the alphabet.
- Count arrangements and combinations systematically.
- Decode two-step family relations (father's wife, brother's son).
- Use remainders to find far positions in repeating patterns.
- Solve read-carefully riddles and digit-counting questions.

Instructions

1. This worksheet has 20 multiple-choice questions. Each question has exactly one correct answer.
2. Read every question carefully. Look at any picture, table or figure before choosing.
3. Each correct answer earns +1 mark. There is no negative marking, so attempt every question.
4. Write your chosen option — (A), (B), (C) or (D) — clearly in the answer box or on a separate sheet.
5. Do all rough work in the margin or on rough paper. Do not use a calculator.
6. After finishing, check your answers against the Answer Key, then read the Detailed Solutions to understand the reasoning behind every question — that is where the real learning happens.

Marking scheme: +1 mark for each correct answer · No negative marking · Maximum 20 marks.



SECTION A · Multiple Choice Questions · 20 Questions

- Q1.** Which letter is exactly MIDWAY between **C** and **K** in the alphabet?
(A) F (B) G
(C) H (D) E
- Q2.** How many letters are there between **D** and **M** in the alphabet (not counting D and M)?
(A) 7 (B) 8
(C) 9 (D) 10
- Q3.** If the alphabet is written BACKWARDS (Z first), which letter comes 5th?
(A) U (B) V
(C) W (D) T
- Q4.** How many different 2-digit numbers can be made using the digits **3** and **7**, using each digit only once per number?
(A) 1 (B) 2
(C) 3 (D) 4
- Q5.** Riya has a red shirt and a blue shirt, and black pants and white pants. How many DIFFERENT shirt-and-pants outfits can she make?
(A) 2 (B) 3
(C) 4 (D) 6
- Q6.** From the word **SCHOOL**, how many DIFFERENT letters are used?
(A) 4 (B) 5
(C) 6 (D) 3
- Q7.** Pointing to a woman, Dev says, 'She is my father's wife.' Who is the woman to Dev?
(A) His aunt (B) His sister
(C) His mother (D) His grandmother
- Q8.** Pointing to a boy, Lata says, 'He is the son of my brother.' Who is the boy to Lata?
(A) Her son (B) Her nephew
(C) Her cousin (D) Her brother
- Q9.** Using the digits **1, 2, 3** once each, how many different 3-digit numbers can be formed?
(A) 3 (B) 4
(C) 6 (D) 9
- Q10.** If every SECOND letter of **ELEPHANT** is removed (the 2nd, 4th, 6th, 8th), which letters REMAIN?
(A) E, E, H, N (B) L, P, A, T
(C) E, E, H, A (D) E, L, P, N



Q11. In a family: A is B's father. B is C's mother. How is A related to C?

- (A) Father (B) Uncle
(C) Grandfather (D) Brother

Q12. How many TRIANGLES of any size are needed to cover a square completely if the square is cut along ONE diagonal?

- (A) 1 (B) 2
(C) 3 (D) 4

Q13. A coin is flipped twice. Which list shows ALL the different possible results (H = heads, T = tails)?

- (A) HH, TT (B) HH, HT, TT
(C) HH, HT, TH, TT (D) HT, TH

Q14. Which word can be formed using ONLY the letters of **TEACHER**?

- (A) CHEAT (B) TIGER
(C) CHAIR (D) REACH (using R twice)

Q15. If 1st March is a Friday, how many SUNDAYS are there in March (31 days)?

- (A) 4 (B) 5
(C) 6 (D) 3

Q16. Aarav, Bina and Chand sit in a row of 3 chairs. Aarav refuses to sit in the middle. In how many ways can the three sit?

- (A) 2 (B) 3
(C) 4 (D) 6

Q17. What is the 100th letter in the repeating pattern **A B C A B C A B C ...** ?

- (A) A (B) B
(C) C (D) Cannot say

Q18. A photo shows Mohan standing to the LEFT of Sita, and Sita to the LEFT of Ravi. Who is in the MIDDLE?

- (A) Mohan (B) Sita
(C) Ravi (D) Cannot say

Q19. How many times does the digit **7** appear when writing all numbers from 70 to 79?

- (A) 10 (B) 11
(C) 12 (D) 9

Q20. Two mothers and two daughters went shopping. Each bought one dress, yet only THREE dresses were bought. How is this possible?

- (A) One dress was torn (B) They were grandmother, mother and daughter
(C) One person bought nothing (D) It is impossible



ANSWER KEY & DETAILED SOLUTIONS

Quick Answer Key

Q	1	2	3	4	5	6	7	8	9	10
Ans	B	B	B	B	C	B	C	B	C	A
Q	11	12	13	14	15	16	17	18	19	20
Ans	C	B	C	A	B	C	A	B	B	B

Step-by-Step Solutions

Q1. Correct answer: (B)

C(3) to K(11): midway value = $(3 + 11) \div 2 = 7 \rightarrow$ the 7th letter is **G**.

Q2. Correct answer: (B)

D(4) and M(13): letters between = $13 - 4 - 1 = 8$ (E, F, G, H, I, J, K, L).

Q3. Correct answer: (B)

Backwards: Z(1), Y(2), X(3), W(4), **V(5)**.

Q4. Correct answer: (B)

Only **37** and **73** — that is 2 numbers.

Q5. Correct answer: (C)

Each of 2 shirts pairs with each of 2 pants: $2 \times 2 = 4$ outfits.

Q6. Correct answer: (B)

S, C, H, O, L — the O repeats. Different letters = 5.

Q7. Correct answer: (C)

Father's wife = **mother**.

Q8. Correct answer: (B)

Brother's son = **nephew**.

Q9. Correct answer: (C)

123, 132, 213, 231, 312, 321 $\rightarrow$ 6 numbers (3 choices $\times$ 2 $\times$ 1).

Q10. Correct answer: (A)

ELEPHANT = E(1) L(2) E(3) P(4) H(5) A(6) N(7) T(8). Removing positions 2, 4, 6, 8 leaves **E, E, H, N**.

Q11. Correct answer: (C)

A is the father of C's mother — that makes A the **grandfather** of C.

Q12. Correct answer: (B)

One diagonal cuts a square into exactly **2 triangles**.

Q13. Correct answer: (C)

Each flip has 2 results, so $2 \times 2 = 4$ outcomes: **HH, HT, TH, TT**.

Q14. Correct answer: (A)

CHEAT uses C, H, E, A, T — all available in TEACHER. (TIGER needs I and G; CHAIR needs I.)

Q15. Correct answer: (B)

1st = Friday $\rightarrow$ 3rd is the first Sunday. Sundays: 3, 10, 17, 24, 31 $\rightarrow$ **5 Sundays**.

**Q16. Correct answer: (C)**

Total ways for 3 people = 6. Ways with Aarav in the middle = 2 (Bina-Aarav-Chand, Chand-Aarav-Bina). Allowed ways = $6 - 2 = 4$.

Q17. Correct answer: (A)

The pattern repeats every 3. $100 \div 3 = 33$ remainder 1, so the 100th letter is the 1st of a group: **A**.

Q18. Correct answer: (B)

Order: Mohan, Sita, Ravi. The middle person is **Sita**.

Q19. Correct answer: (B)

Each of 70–79 has a 7 in the tens place (10 times), and 77 has one more in the ones place. Total = $10 + 1 = 11$.

Q20. Correct answer: (B)

A grandmother, her daughter, and her granddaughter are 'two mothers and two daughters' but only **three people** — so three dresses.

